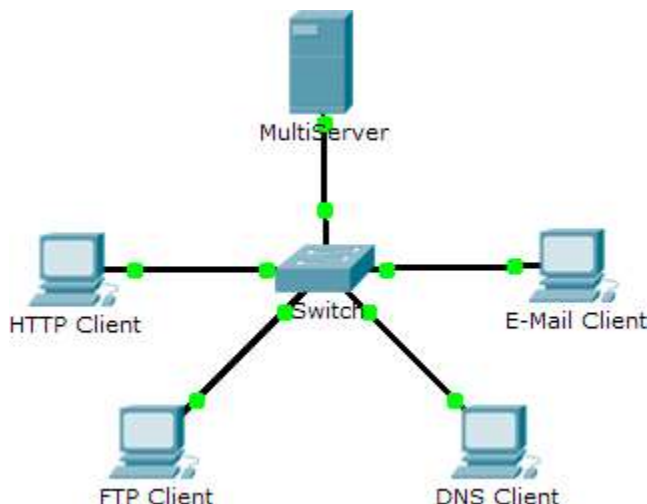


Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.
- Close the **MultiServer** window.

Step 2: Generate web (HTTP) traffic.

- a. Switch to Simulation mode.
- b. Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- d. Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

- a. Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- c. Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

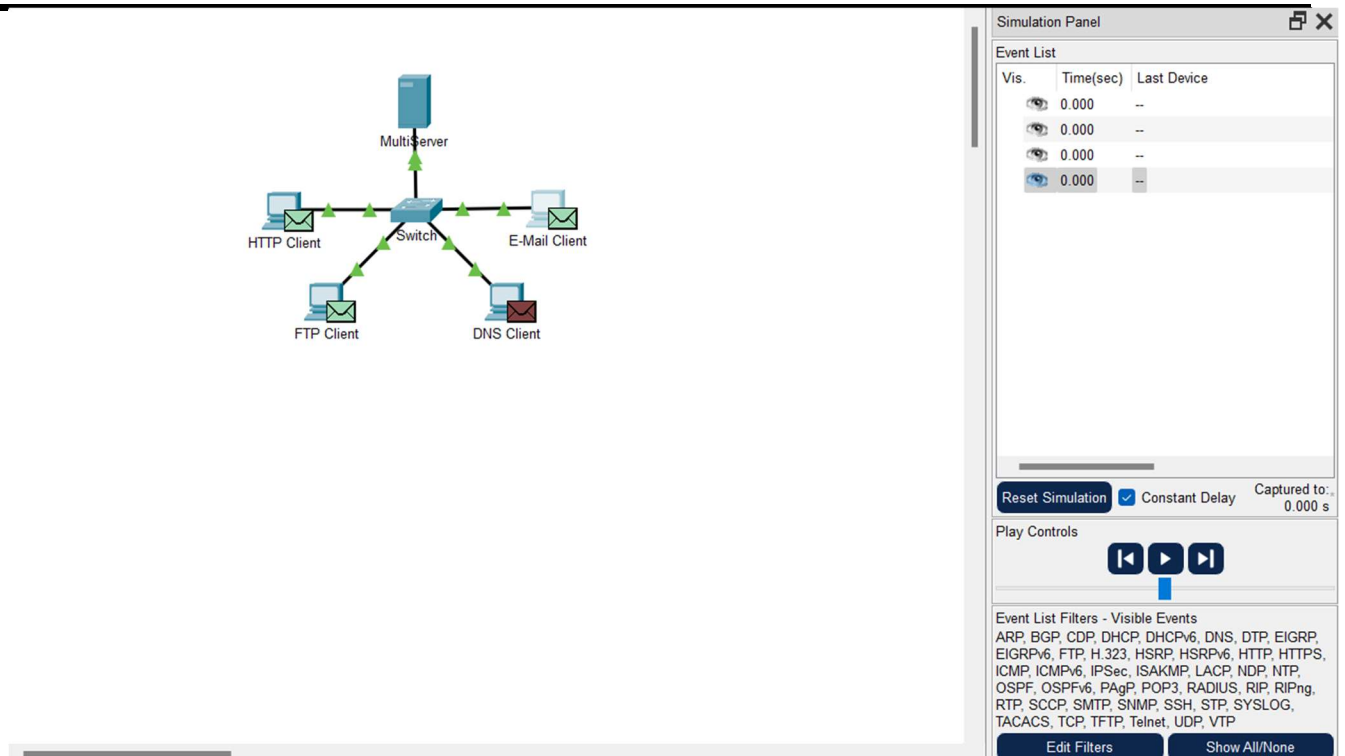
- a. Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- c. Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

- a. Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- b. Click **Compose** and enter the following information:
 - 1) **To:** user@multiserver.pt.ptu
 - 2) **Subject:** Personalize the subject line
 - 3) **E-Mail Body:** Personalize the Email
- c. Click **Send**.
- d. Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.



Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--
	0.000	--

Reset Simulation ☒ Constant Delay Captured to: 0.000 s

Play Controls

Event List Filters - Visible Events

ARP, BGP, CDP, DHCP, DHCPv6, DNS, DTP, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, LACP, NDP, NTP, OSPF, OSPFv6, PAgP, POP3, RADIUS, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, VTP

Edit Filters Show All/None

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

- Click **Capture/Forward** once. All of the PDUs are transferred to the switch.



Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--
	0.000	--
	0.001	HTTP Client

- Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?



Simulation Panel

Event List

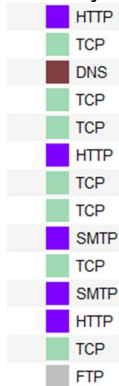
Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--
	0.000	--
	0.001	HTTP Client
	0.001	FTP Client
	0.001	DNS Client
	0.001	E-Mail Client
	0.002	Switch
	0.002	--

The PDUs that disappear have been stored in the switch.

Packet Tracer Simulation - TCP and UDP Communications

- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?
multiplexing

- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are there so many different colors?



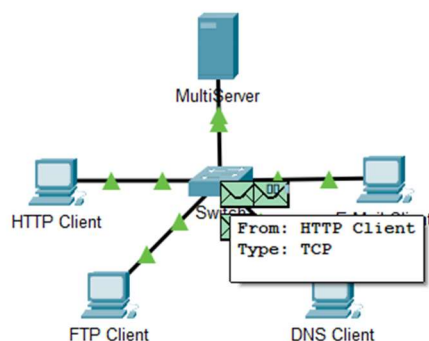
different colors represent different protocols

- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

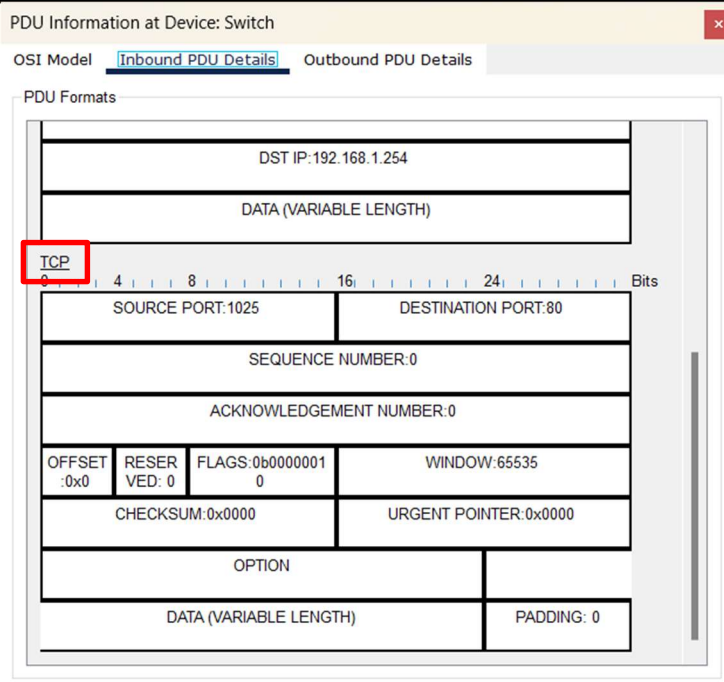
Step 2: Examine HTTP traffic as the clients communicate with the server.

- a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
- 1) Click **Edit Filters** and toggle the **Show All/None** check box.
 - 2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.
- b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.



- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

Packet Tracer Simulation - TCP and UDP Communications

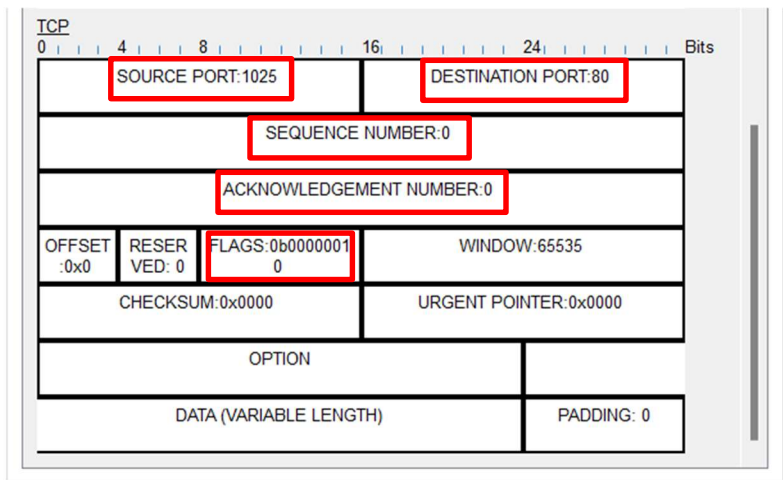


TCP

Are these communications considered to be reliable?

Yes. TCP is a reliable transport which has a connection oriented between sending and receiving process.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?



SRC PORT: 1025

DEST PORT: 80

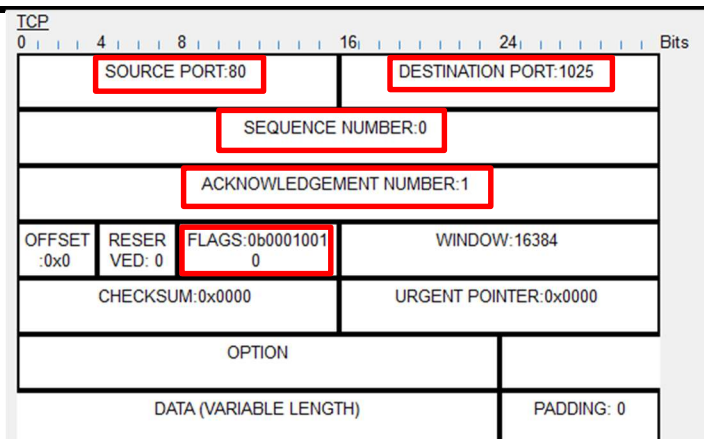
SEQUENCE NUM: 0

ACK NUM: 0

FLAG = SYN

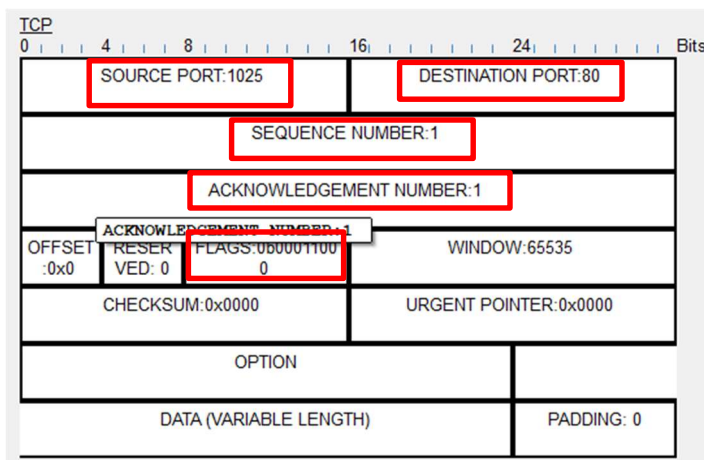
- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Packet Tracer Simulation - TCP and UDP Communications



SRC PORT: 80 (reversed)
 DEST PORT: 1025 (reversed)
 SEQUENCE NUM: 0 (unchanged)
 ACK NUM: 1 (changed)
 FLAGS = SYN+ ACK (changed)

- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.
- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

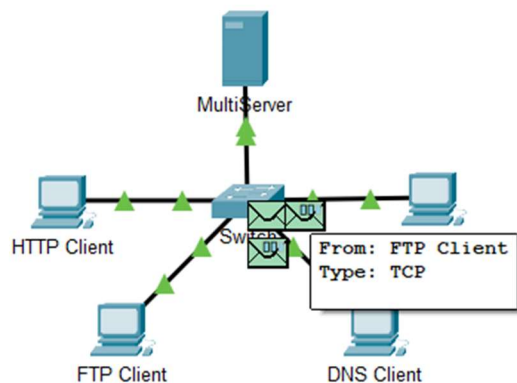


SRC PORT: 1025 (same as the first PDU)
 DEST PORT: 80 (same as the first PDU)
 SEQUENCE NUM: 1 (changed)
 ACK NUM: 1 (unchanged)
 FLAGS = PSH + ACK

- i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.



- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

PDU Information at Device: Switch

OSI Model **Inbound PDU Details** Outbound PDU Details

PDU Formats

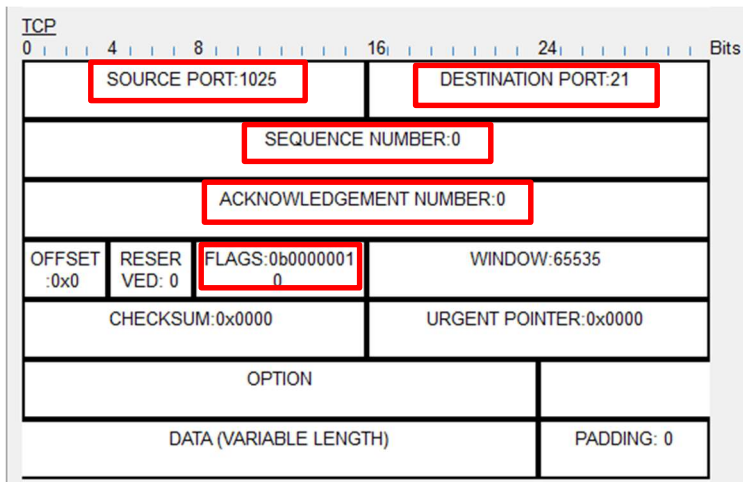
DST IP:192.168.1.254			
DATA (VARIABLE LENGTH)			
TCP			
SOURCE PORT:1025		DESTINATION PORT:21	
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:0			
OFFSET:0x0	RESERVED:0	FLAGS:0b00000010	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

TCP

Are these communications considered to be reliable?

Yes. TCP is a reliable transport which has a connection oriented between sending and receiving process.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?



SRC PORT: 1025

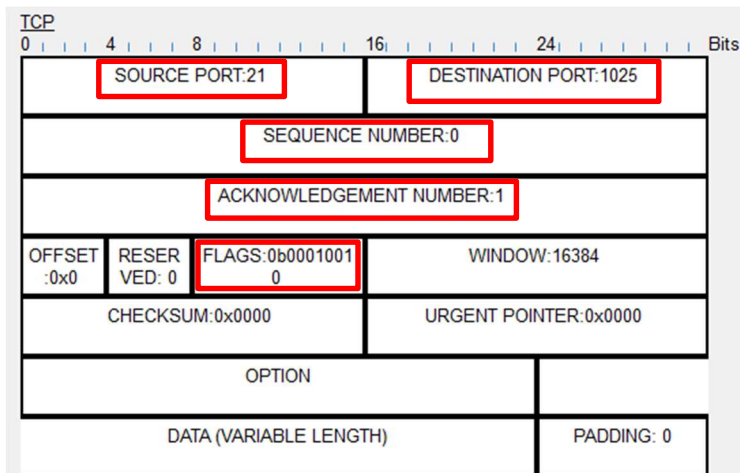
DEST PORT: 21

SEQUENCE NUM: 0

ACK NUM: 0

FLAGS = SYN

- Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.
- Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?



SRC PORT: 21 (reversed)

DEST PORT: 1025 (reversed)

SEQUENCE NUM: 0 (unchanged)

ACK NUM: 1 (changed)

FLAGS = SYN + ACK

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

Packet Tracer Simulation - TCP and UDP Communications

TCP																Bits																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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SRC PORT: 1025 (same as the first PDU)

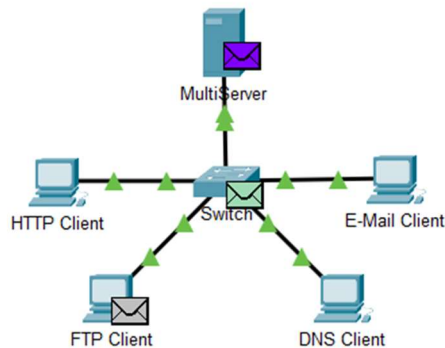
DEST PORT: 21 (same as the first PDU)

SEQUENCE NUM: 1 (changed)

ACK NUM: 1 (unchanged)

FLAGS = ACK

- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.



- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

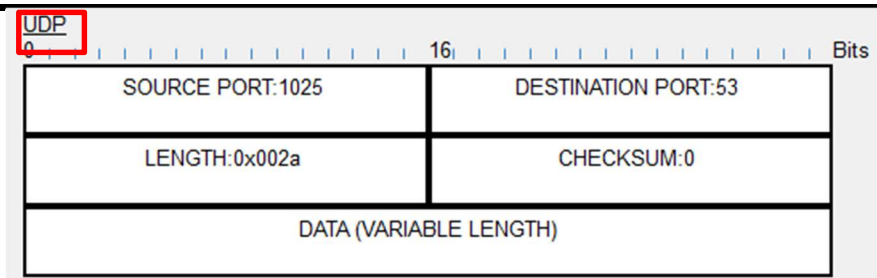
FTP Response																Bytes															
Code:220																															
Message:Welcome to PT Ftp server																															

"Welcome to PT Ftp server"

- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.
- Click the PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

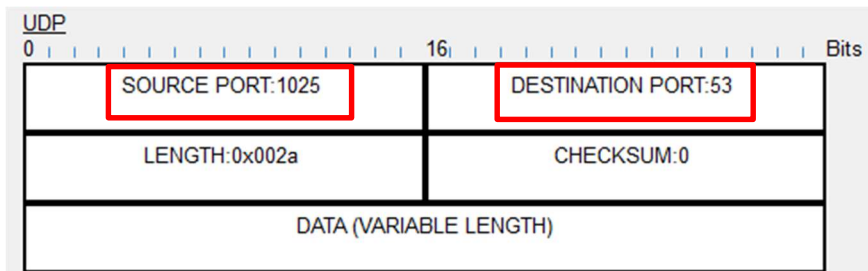


UDP

Are these communications considered to be reliable?

No, UDP is unreliable data transfer between sending and receiving process because it does not have a connection oriented.

- d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

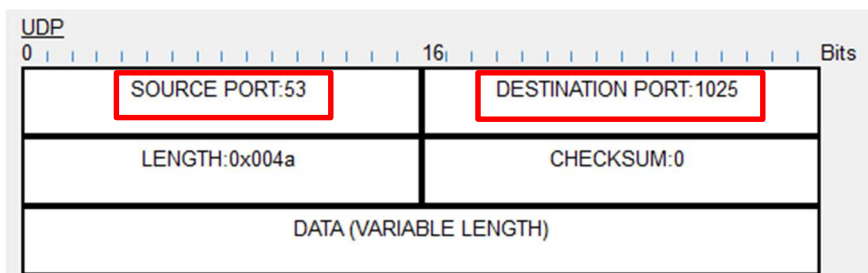


SRC PORT: 1025

DEST PORT: 53

There is no SEQUENCE NUMBER and ACK NUM because UDP does not need to establish a reliable connection.

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.
f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?



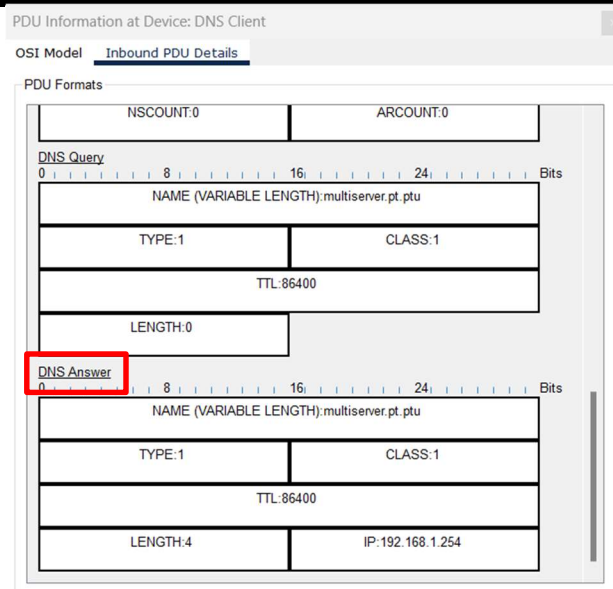
SRC PORT: 53 (reversed)

DEST PORT: 1025 (reversed)

There is not any SEQUENCE NUM and ACK NUM show in UDP connection.

- g. What is the last section of the **PDU** called?

Packet Tracer Simulation - TCP and UDP Communications



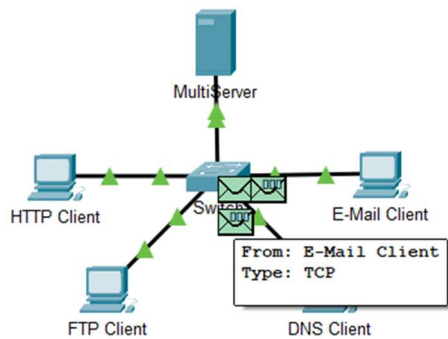
DNS Answer

- h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **POP3**, **SMTP** and **TCP**.

- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.



- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

PDU Information at Device: Switch

OSI Model **Inbound PDU Details** Outbound PDU Details

PDU Formats

DST IP: 192.168.1.254			
DATA (VARIABLE LENGTH)			
TCP			
SOURCE PORT: 1025		DESTINATION PORT: 25	
SEQUENCE NUMBER: 0			
ACKNOWLEDGEMENT NUMBER: 0			
OFFSET: 0x0	RESERVED: 0	FLAGS: 0b00000010	WINDOW: 65535
CHECKSUM: 0x0000		URGENT POINTER: 0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

TCP

Are these communications considered to be reliable?

Yes. TCP is a reliable transport which has a connection oriented between sending and receiving process.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Packet Tracer Simulation - TCP and UDP Communications

TCP																Bits																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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SRC PORT: 1025

DEST PORT: 25

SEQUENCE NUM: 0

ACK NUM: 0

FLAGS = SYN

- Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.
- Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

TCP																																Bits																																																							
0																4																								8																16																24															
SOURCE PORT:25																								DESTINATION PORT:1025																																																															
SEQUENCE NUMBER:0																																																																																							
ACKNOWLEDGEMENT NUMBER:1																																																																																							
OFFSET:0x0								RESERVED:0								FLAGS:0b00010010								WINDOW:16384																																																															
CHECKSUM:0x0000																								URGENT POINTER:0x0000																																																															
OPTION																																																																																							
DATA (VARIABLE LENGTH)																												PADDING: 0																																																											

SRC PORT: 25 (reversed)

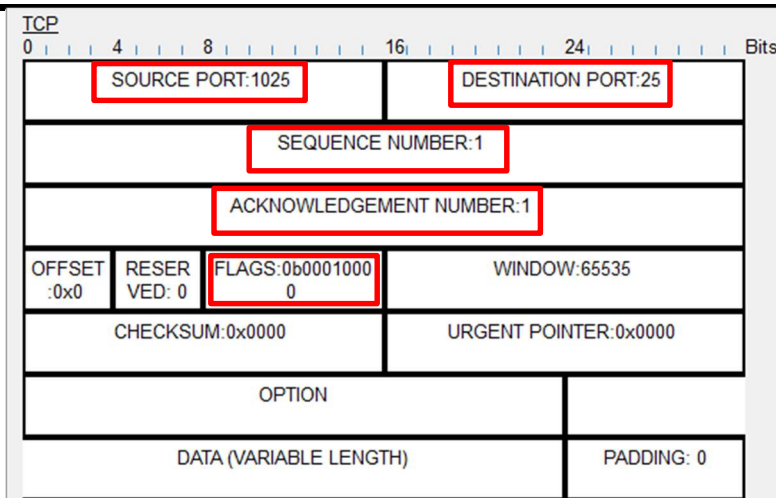
DEST PORT: 1025 (reversed)

SEQUENCE NUM: 0 (unchanged)

ACK NUM: 1 (changed)

FLAGS = SYN + ACK

- Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?



SRC PORT: 1025 (same as the first PDU)

DEST PORT: 25 (same as the first PDU)

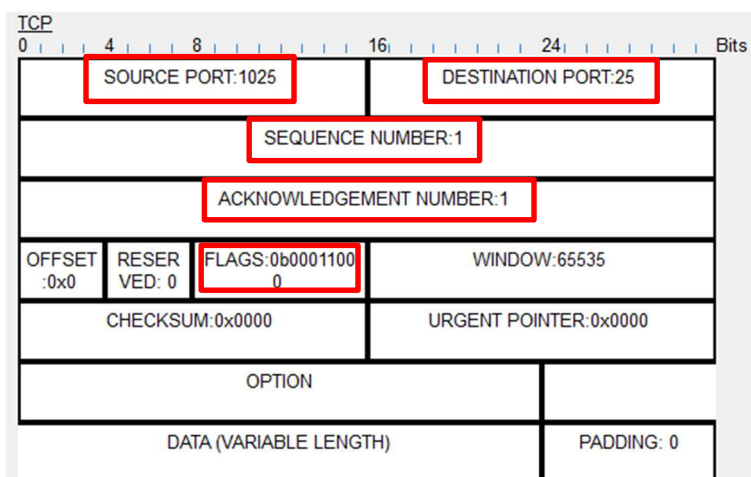
SEQUENCE NUM: 1 (different)

ACK NUM: 1 (unchanged)

FLAGS = ACK

- h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.

- i. How are the port and sequence numbers different from the previous two **PDU**s?



SRC PORT: 1025 (same as the first PDU)

DEST PORT: 25 (same as the first PDU)

SEQUENCE NUM: 1 (different)

ACK NUM: 1 (unchanged)

FLAGS = PSH + ACK

- j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?
SMTP is associated with TCP port 25 and POP3 is associated with TCP port 110.

- k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

Packet Tracer Simulation - TCP and UDP Communications

- a. To see TCP active sessions, perform the following steps in quick succession:
 - 1) Switch back to **Realtime** mode.
 - 2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **netstat** command. What protocols are listed in the left column?

```
C:\>netstat

Active Connections

  Proto Local Address           Foreign Address         State
  TCP   192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
C:\>
```

TCP

What port numbers are being used by the server?

Port 21

- c. What states are the sessions in?
Established
- d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?

```
Active Connections

  Proto Local Address           Foreign Address         State
  TCP   192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
C:\>netstat

Active Connections

  Proto Local Address           Foreign Address         State
  TCP   192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
C:\>netstat

Active Connections

  Proto Local Address           Foreign Address         State
  TCP   192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
C:\>
```

Port 21: FTP

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.1.254
Trying to connect...192.168.1.254
Connected to 192.168.1.254
220- Welcome to PT Ftp server
Username:
```

The server is waiting for the user input.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	